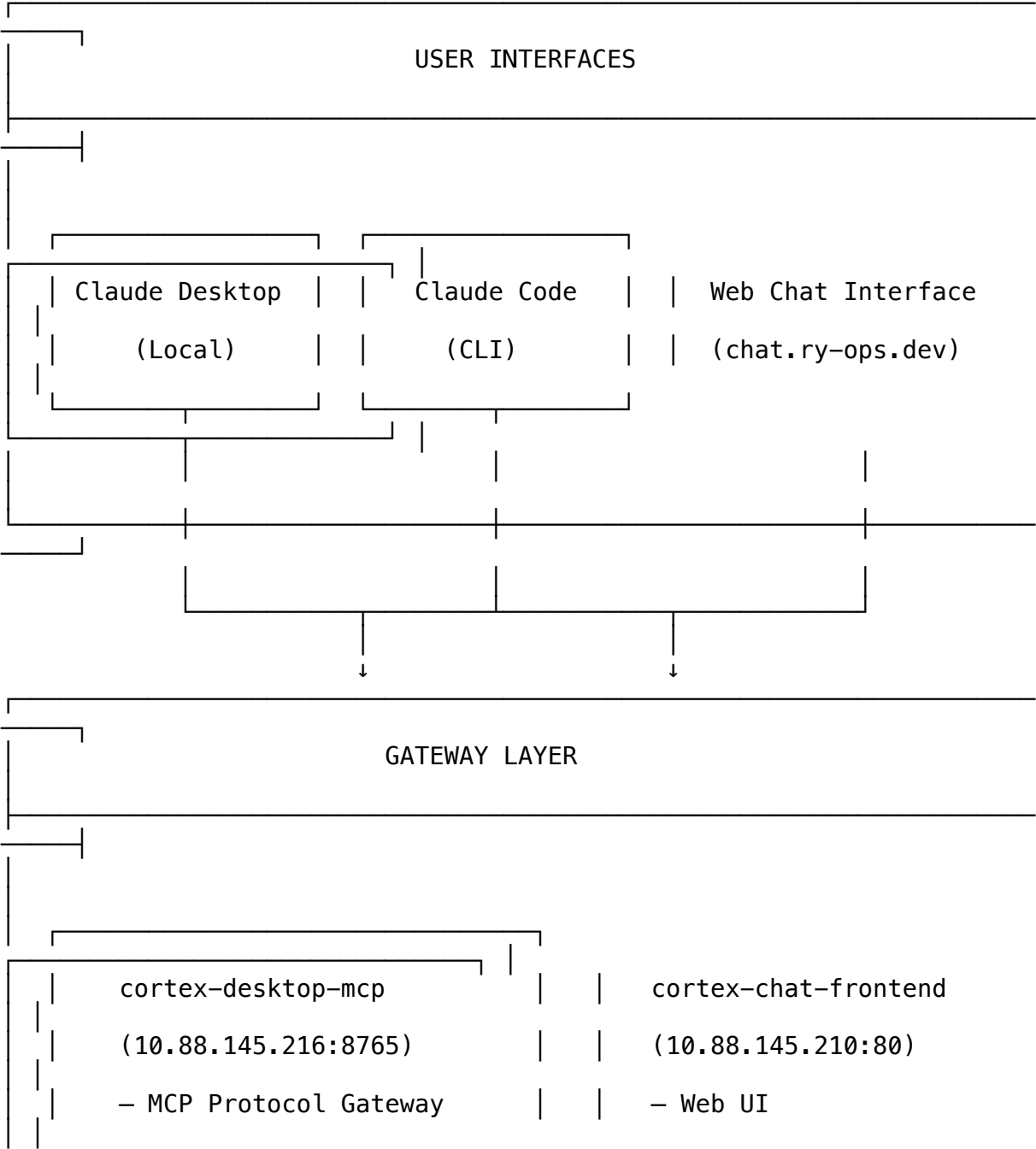


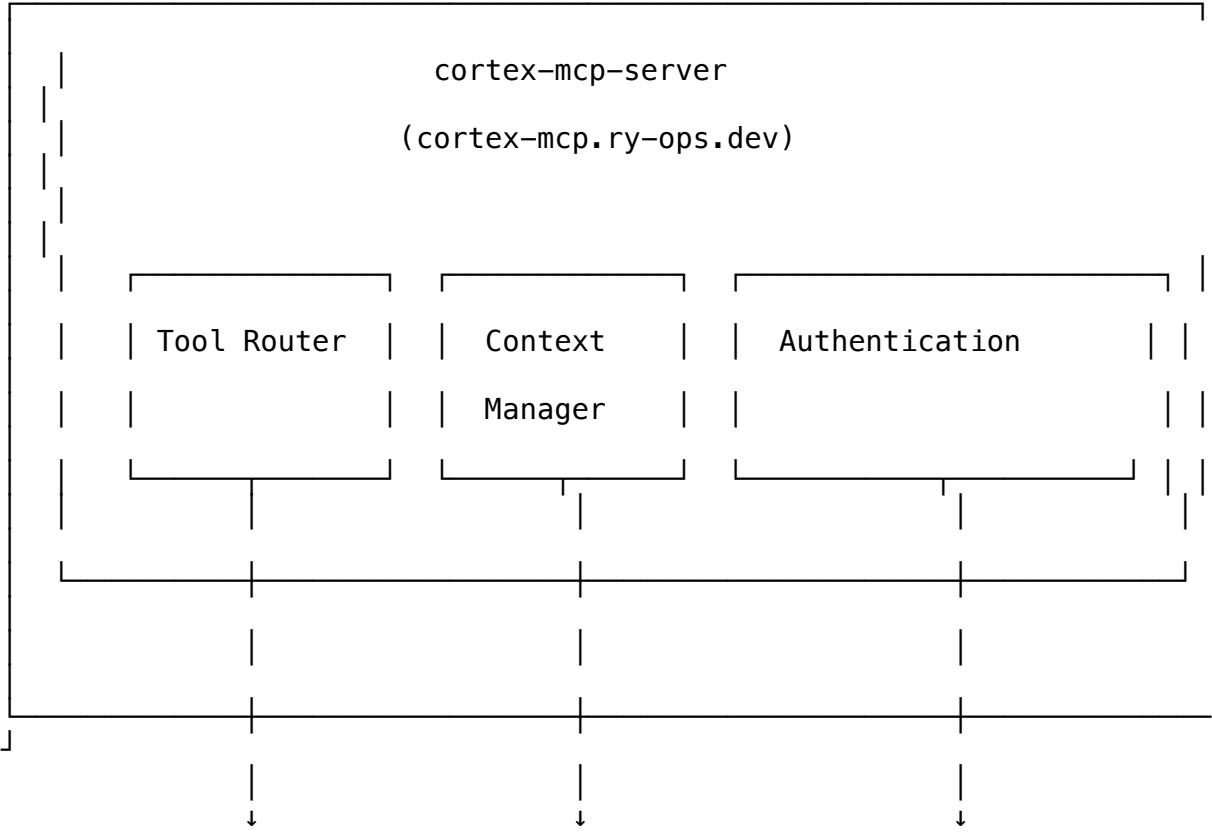
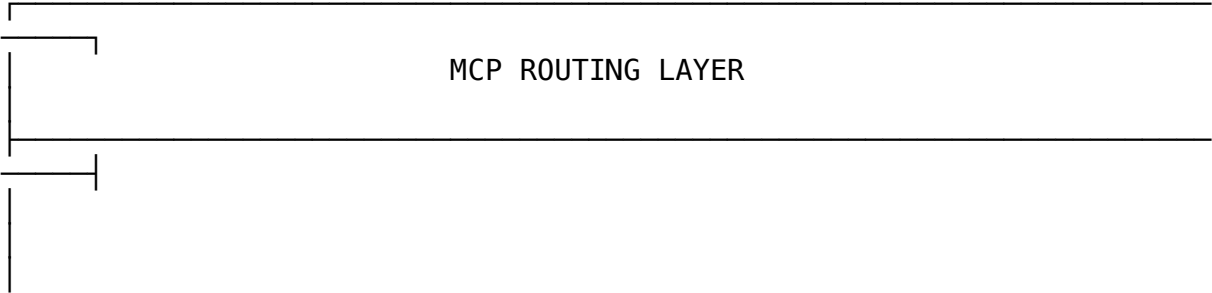
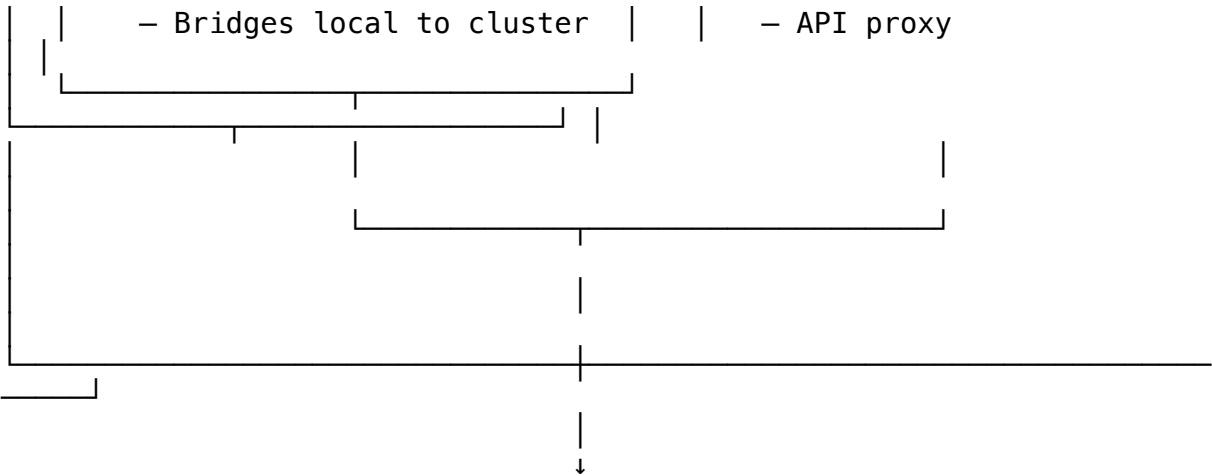
Cortex Visual Flow Diagram

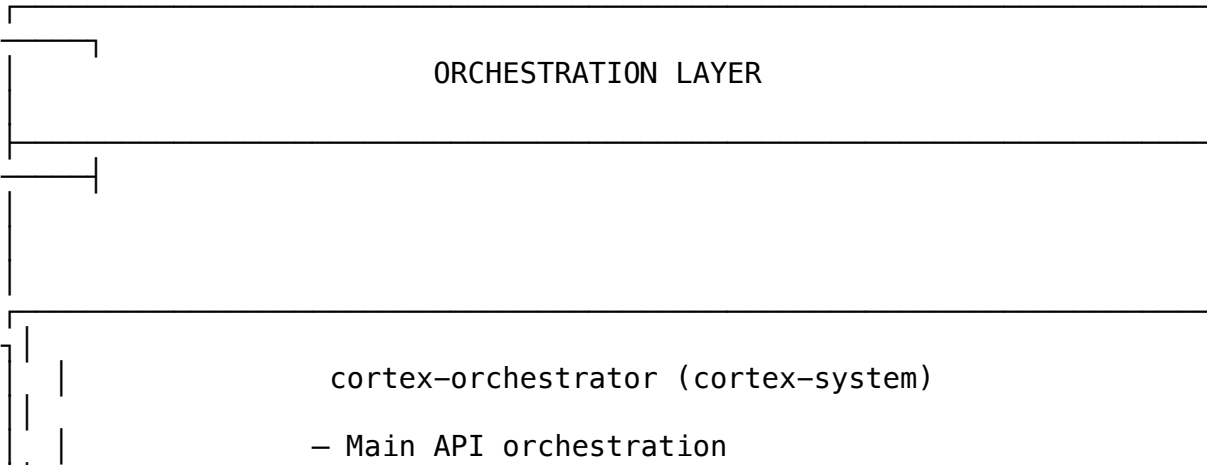
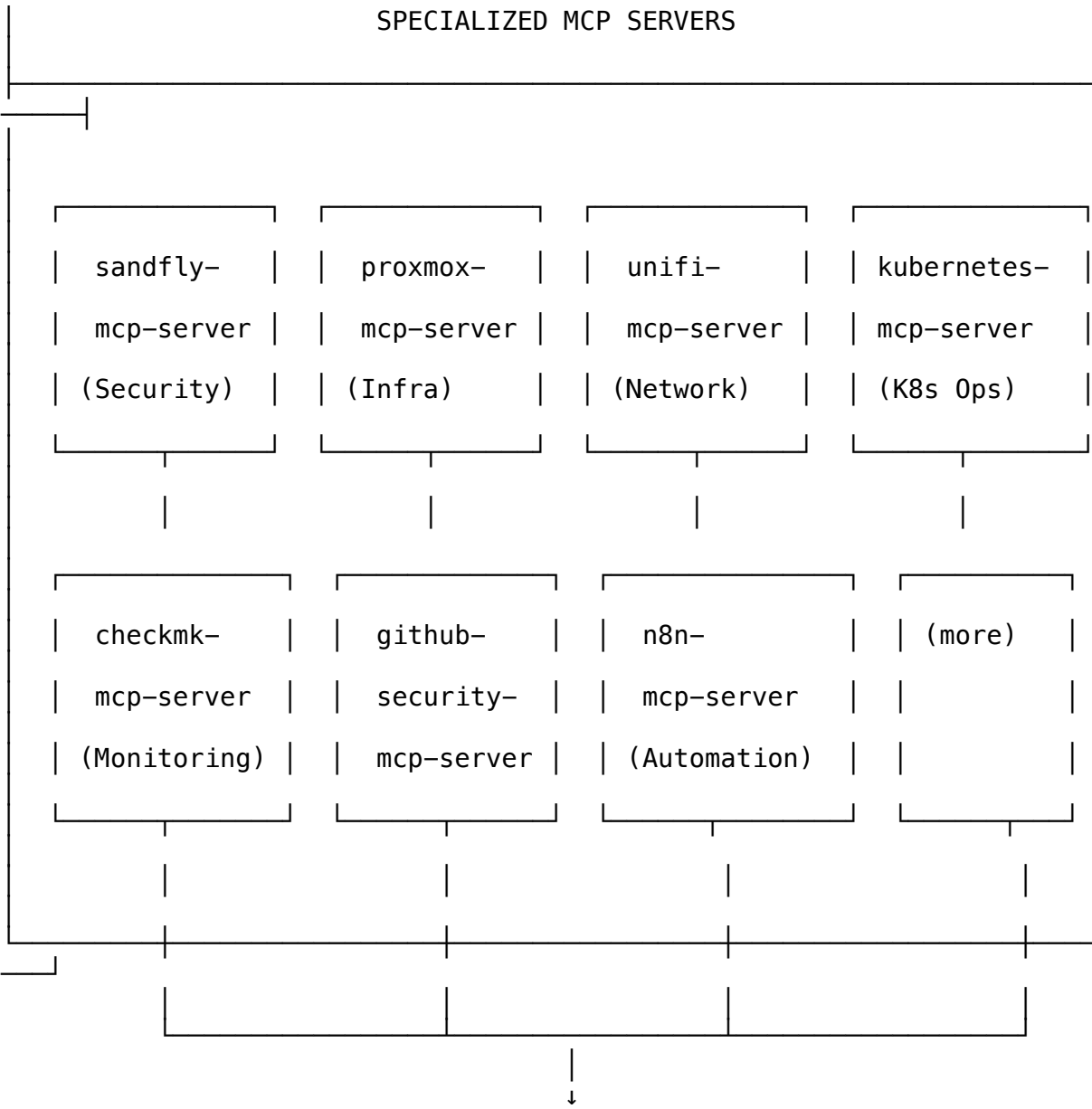
****Complete Data Flow Architecture****
****Date**:** 2026-01-15

High-Level Architecture

...







- Task routing & coordination
- Multi-agent workflow management

↓

redis-master (Task Queue)
cortex:queue:{critical|high|medium|low}

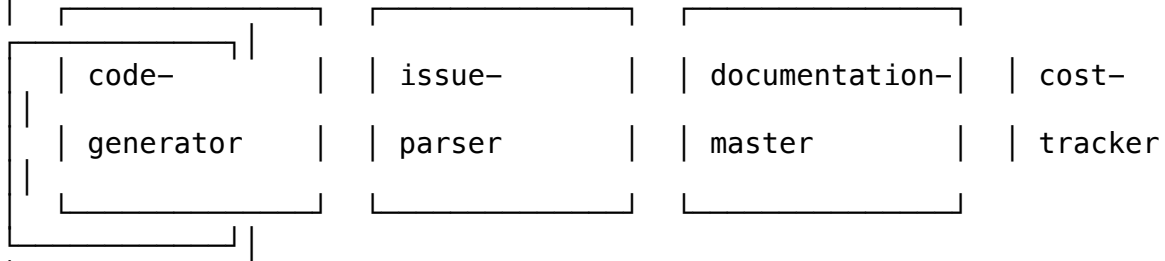
↓

WORKER LAYER

cortex-queue-worker (2-25 replicas)

- Process queued tasks
- HPA scaling based on load

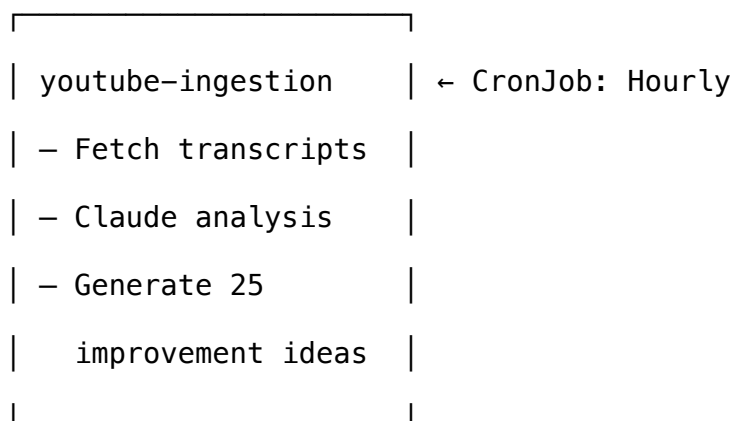
Calls specialized services based on task type:



AUTONOMOUS LEARNING PIPELINE (Cortex Online School)

YouTube Videos

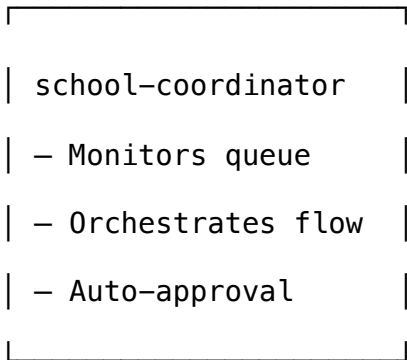
↓



↓

11

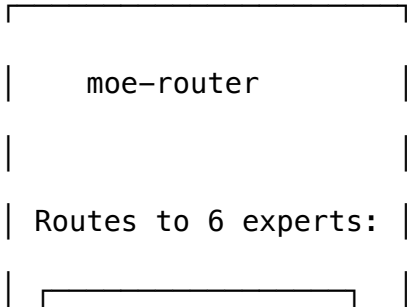
11



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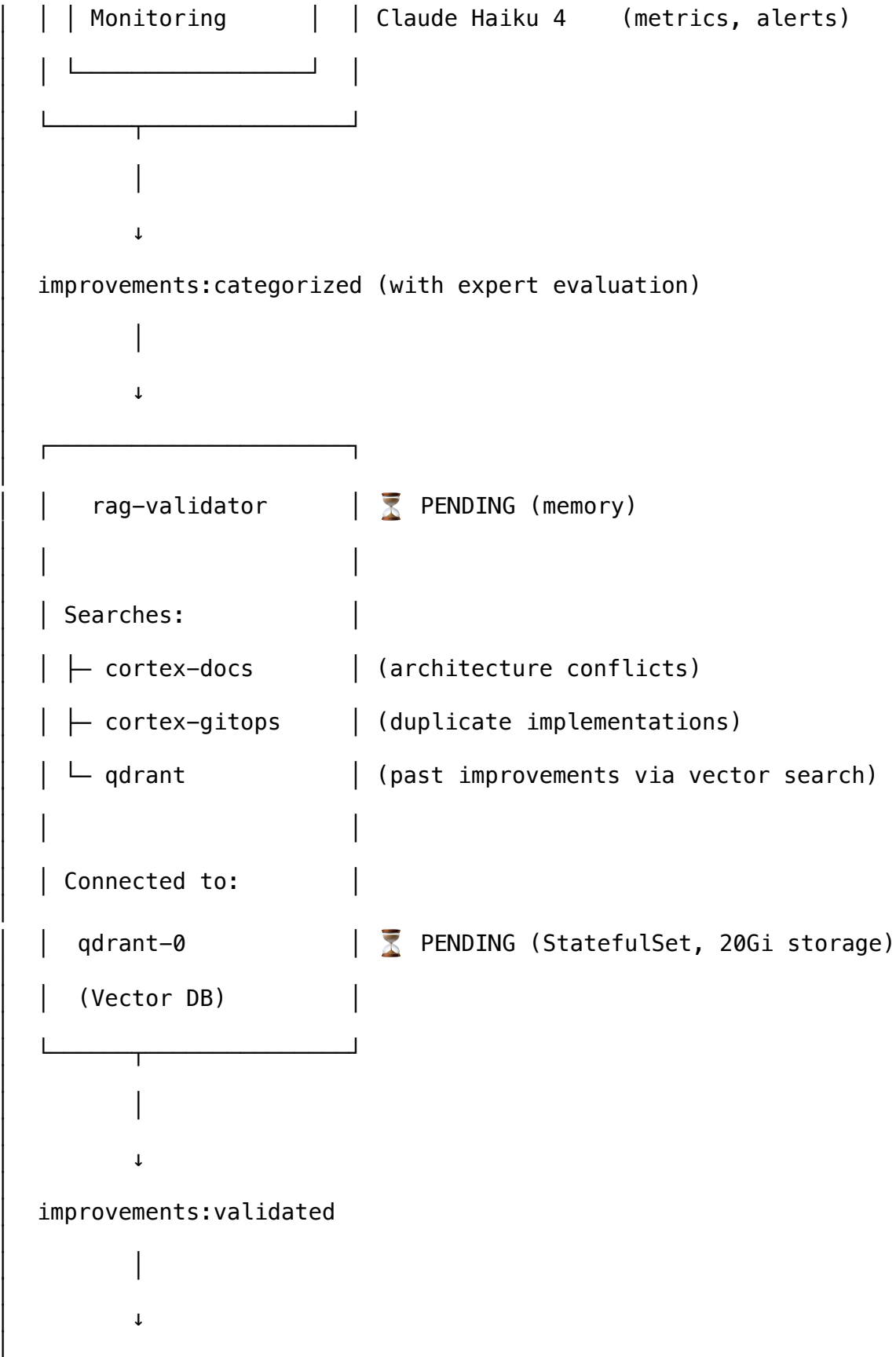
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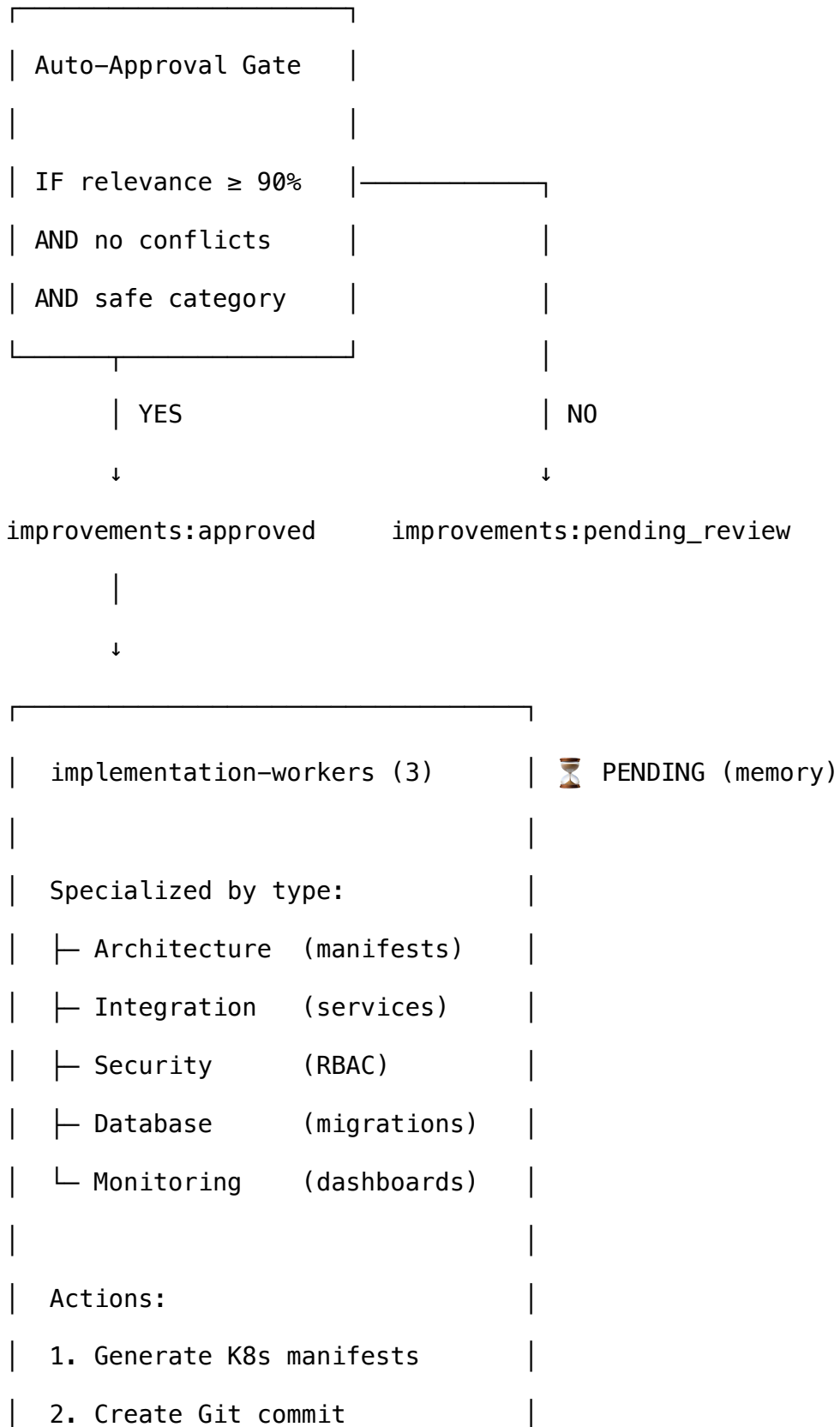
1

Cla

1

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3. Push to cortex-gitops



GitHub: cortex-gitops

- New commit with manifests

- Detailed commit message

- Co-authored by Cortex School



ArgoCD

- Polls GitHub every 3 min

- Detects new commit

- Auto-syncs to cluster

- Self-heals drift



Kubernetes Deployment

- New pods/services created

- Resources updated

| _____ |

|

↓

improvements:deployed

|

↓

| _____ |

| health-monitor |  RUNNING

| |

| Monitors for 5 minutes: |

| └─ Pod status (Running/Ready) |

| └─ Readiness probes |

| └─ Prometheus metrics |

| | (error rate, latency) |

| └─ Logs (ERROR/FATAL/panic) |

| └─ Dependency connectivity |

| |

| IF ALL PASS: |

| → improvements:verified  |

| |

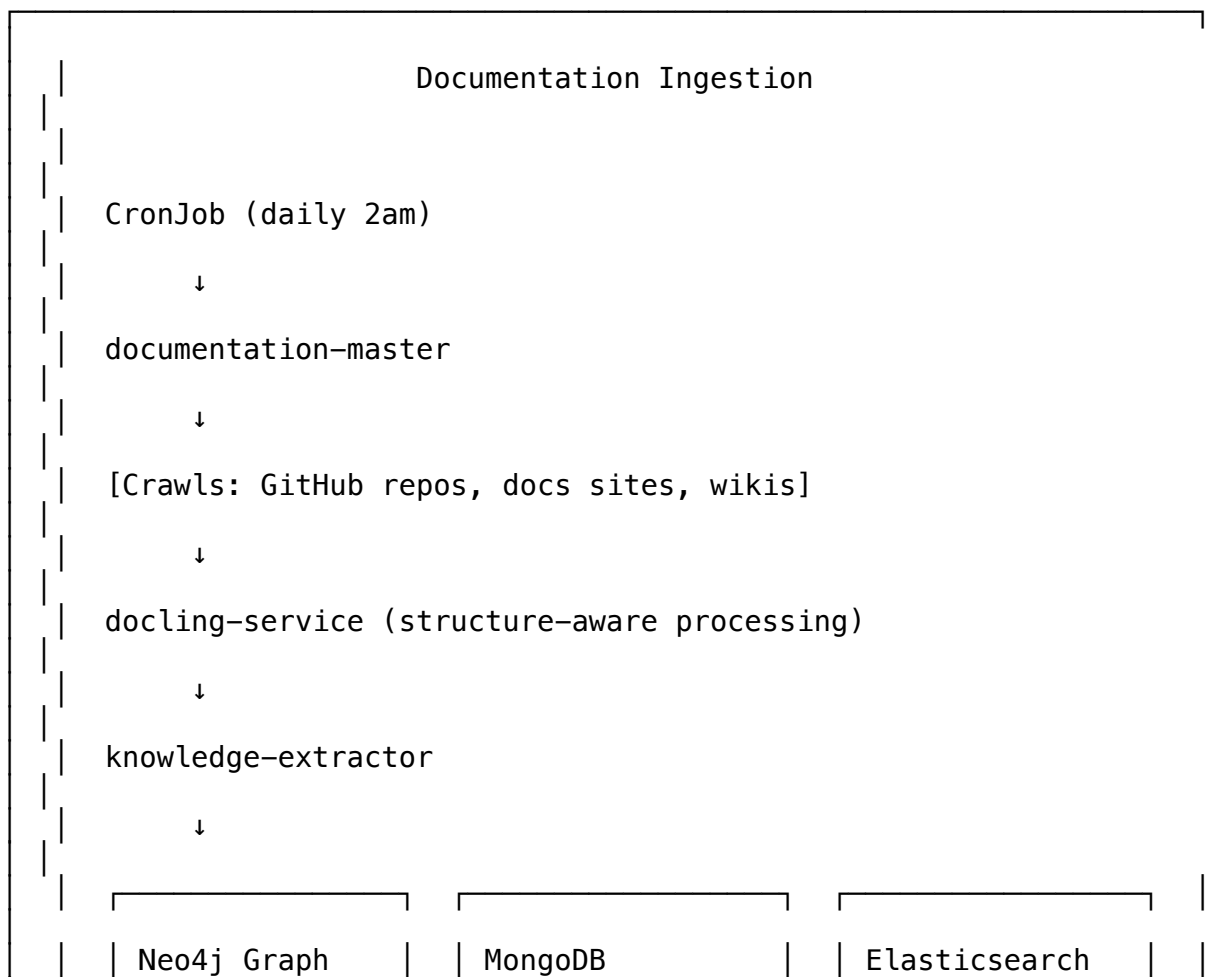
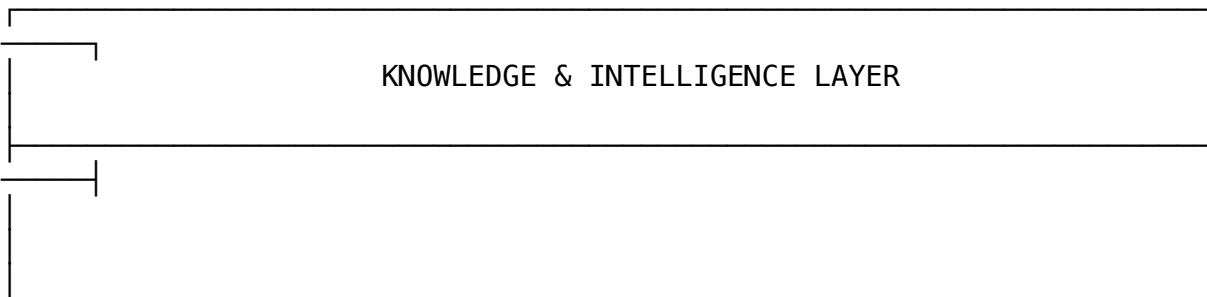
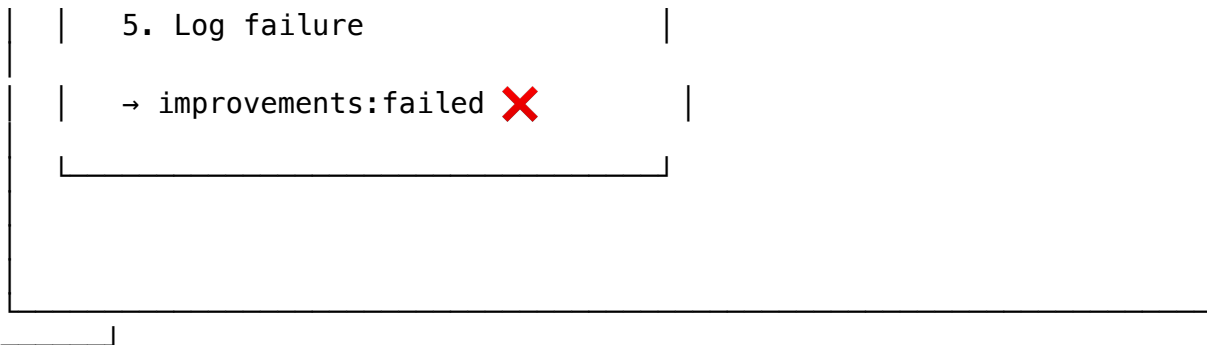
| IF ANY FAIL: |

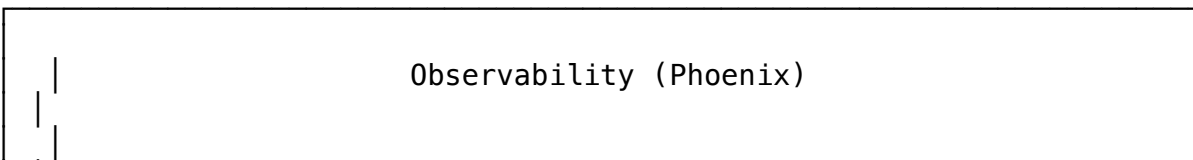
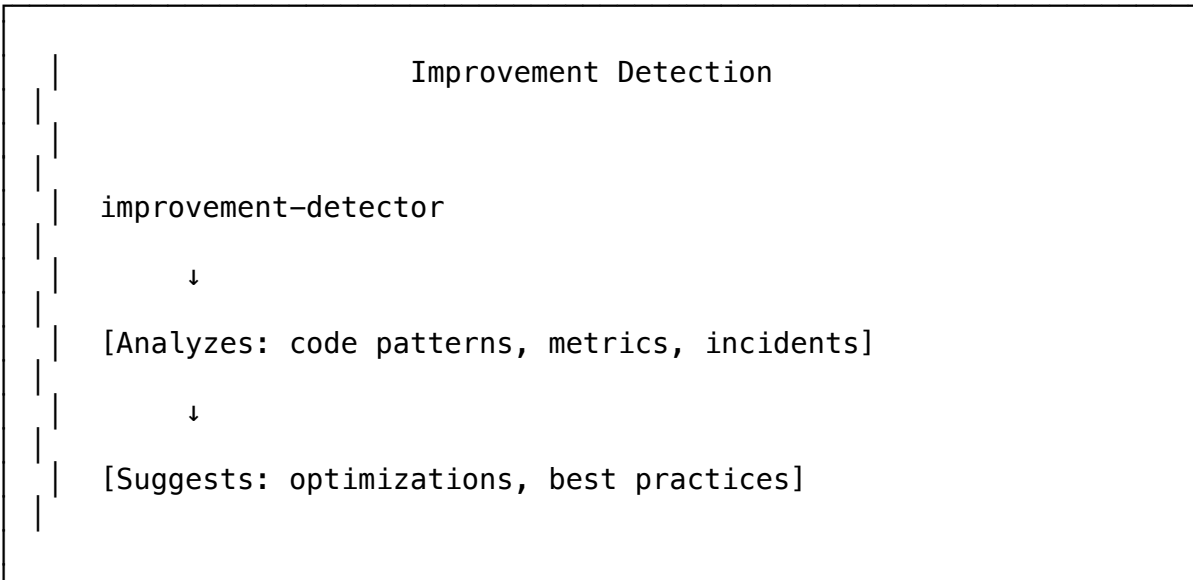
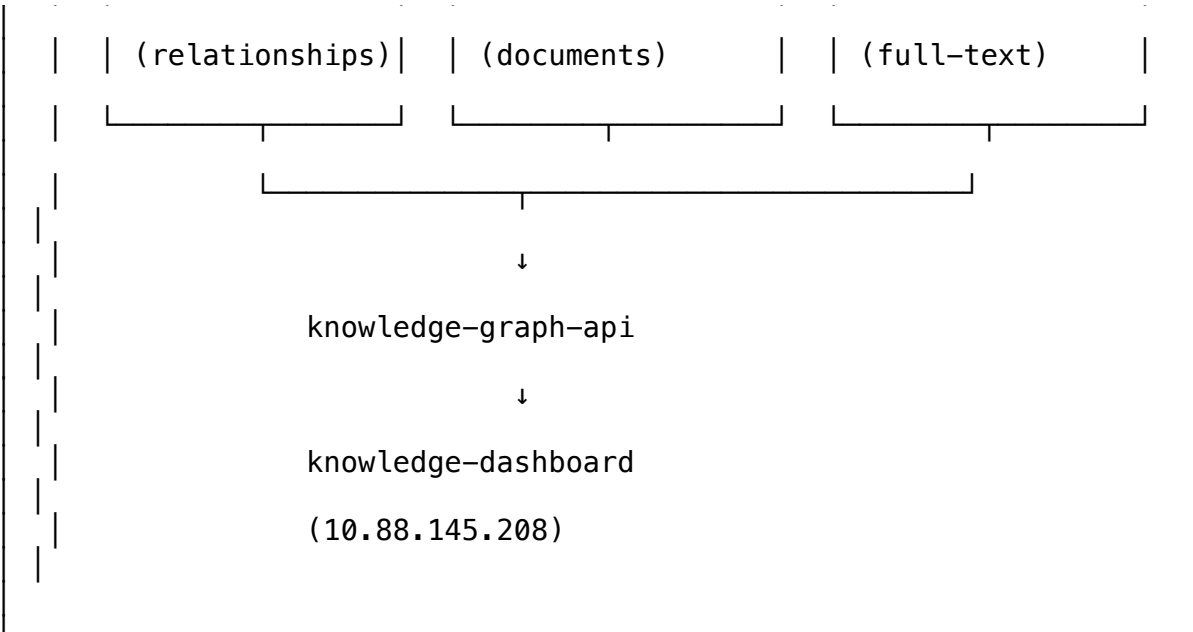
| 1. git revert <commit> |

| 2. Push rollback |

| 3. Force ArgoCD sync |

| 4. Verify healthy |





All AI services → OpenTelemetry → phoenix

Dashboard: observability.ry-ops.dev

- LLM traces
- Token usage
- Latency metrics
- Cost tracking

SECURITY & MONITORING LAYER

Runtime Security (Falco)

All K8s nodes → Falco DaemonSet

↓

[Detects: syscalls, file access, network anomalies]

↓

falco-falcosidekick (alert forwarding)

↓

falco-falcosidekick-ui (dashboard)

Vulnerability Scanning (Trivy)

CronJob (daily 2am)

↓

trivy-cluster-scan

↓

[Scans: containers, configs, secrets]

↓

[Reports: CVEs, misconfigurations]

EDR (Sandfly)

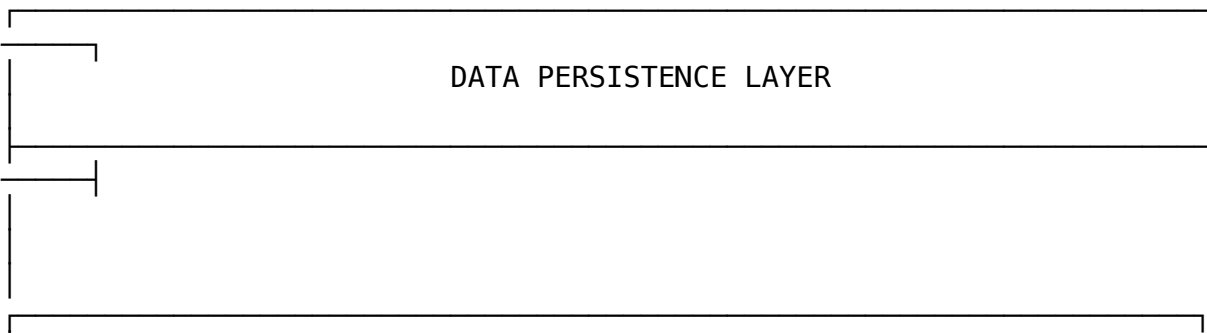
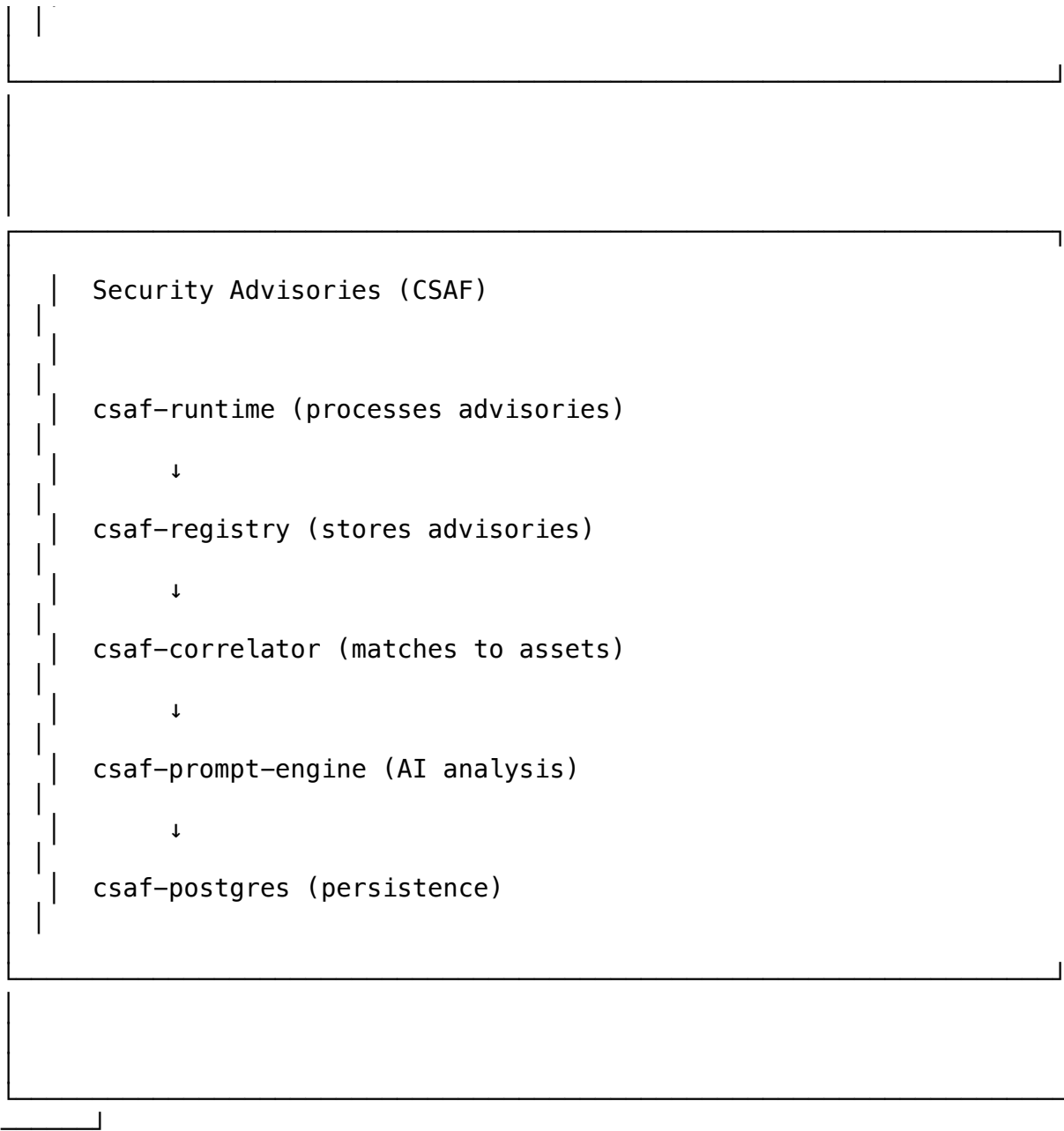
sandfly-mcp-server → Sandfly API (10.88.140.176)

↓

[Monitors: hosts, processes, network]

↓

sandfly-web (sandfly.ry-ops.dev)



Primary Database (PostgreSQL)

postgres-postgresql (StatefulSet 1/1)

- ClusterIP: 10.43.135.206:5432
- Longhorn persistent volume
- Daily backups (CronJob 2am)
- PostgreSQL exporter (metrics)
- pgAdmin (admin UI)

Used by:

- └ cortex-orchestrator
- └ csaf-postgres
- └ Most application services
- └ Knowledge graph metadata

Cache & Queues (Redis)

redis-master (StatefulSet 1/1)

- ClusterIP: 10.43.48.239:6379
- Persistent volume

redis-queue (cortex namespace)

- ClusterIP: 10.43.10.63:6379

- Task queues

- YouTube improvement pipeline

Multiple Redis instances per namespace:

- └ cortex-chat/redis (sessions)

- └ cortex-dev/redis (dev cache)

- └ cortex-csaf/redis (advisory cache)

Storage Classes

Longhorn (distributed block storage)

- Replicated across nodes

- Automatic backups

- Used for: databases, vector stores, logs

GITOPS LAYER

Source of Truth (GitHub)

cortex-gitops

- 120+ Kubernetes manifests
- 7 ArgoCD Applications
- 23 namespaces

↓

cortex-platform

- Application source code
- Service implementations
- 10,661 files

↓

cortex-docs

- Architecture documentation
- Design decisions
- API specs

Continuous Deployment (ArgoCD)

Polls GitHub every 3 minutes

↓

Detects changes in cortex-gitops

↓

Auto-syncs to cluster

- Self-heal: true (reverts manual changes)

- Prune: true (deletes removed resources)

- Auto-sync: true

↓

kubectl apply -f <manifests>

↓

Kubernetes reconciles state

...

Request Flow Examples

Example 1: User asks Claude to analyze cluster health

```

User: "Show me unhealthy pods in the cluster"

↓

Claude Desktop (local)

↓ (MCP JSON-RPC)

cortex-desktop-mcp (10.88.145.216:8765)

↓

cortex-mcp-server (mcp routing)

↓

kubernetes-mcp-server (specialized tool)

↓

K8s API (kubectl get pods --field-selector=status.phase!=Running)

↓

Returns: List of unhealthy pods

↓

cortex-desktop-mcp

↓

Claude Desktop

↓

User sees formatted response

```

Example 2: User requests code generation

```

User: "Generate a Deployment manifest for nginx"

↓

Claude Desktop / Code

↓

cortex-desktop-mcp

↓

cortex-orchestrator

↓

redis-master (queue task)

↓

cortex-queue-worker (picks up task)

↓

code-generator (specialized service)

↓

[Claude API generates code]

↓

Returns manifest YAML

↓

User receives generated code

```

Example 3: YouTube video triggers autonomous improvement

...

```
YouTube: "Advanced Kubernetes Patterns" video
↓
youtube-ingestion (hourly cron)
↓
[Fetches transcript, analyzes with Claude]
↓
Generates 25 improvements with relevance scores
↓
redis-queue:improvements:raw (sorted set)
↓
school-coordinator (monitors queue)
↓
moe-router (routes to Architecture Expert – Opus 4.5)
↓
Expert evaluates: feasibility=high, impact=medium, priority=high
↓
redis-queue:improvements:categorized
↓
rag-validator
├─ Searches cortex-docs (no conflicts found)
├─ Searches cortex-gitops (not already implemented)
└─ Searches qdrant (no similar improvements)
↓
redis-queue:improvements:validated
↓
Auto-approval gate (relevance 95% ≥ 90% ✅)
↓
redis-queue:improvements:approved
↓
implementation-worker (generates manifest)
↓
Git commit to cortex-gitops
↓
Push to GitHub
↓
ArgoCD detects change (within 3 min)
↓
Auto-syncs to cluster
↓
New Deployment created
↓
redis-queue:improvements:deployed
↓
health-monitor (watches for 5 minutes)
├─ Checks pod status every 10s
├─ Queries Prometheus metrics
├─ Scans logs for errors
└─ Tests dependencies
↓
```

All checks pass 

↓

redis-queue:improvements:verified

↓

Improvement successfully implemented!
\\

****Status****: Comprehensive visual flow as of 2026-01-15

****Note****: Cortex School pipeline partially active (coordinator +
health-monitor running)